Hydropower plants converts the kinetic energy of flowing water into electricity. This is called hydroelectricity.

22.6.6 Tidal Energy

Tidal energy is the **energy obtained from the movement of water due to ocean tides**. Tides are the rise and fall of sea levels caused by the combined effects of the gravitational forces exerted on the oceans of the earth.

A tidal stream is a fast flowing body of water created by tides. Turbines are placed in tidal streams. When the tides hit the turbine, the turbine rotates and converts the tidal energy into electric energy

Advantages of tidal energy

(i) Tidal energy does not produce any pollution.

(ii) It does not use any fuel and does not produce any waste.

(iii) Tides are predictable, so tidal energy can be produced at any time.

(iv) Water is denser than air and therefore can generate electricity at lower speeds than wind turbines.

22.7 Rainwater Harvesting

Rainwater harvesting is a technique of **collecting and storing rainwater** for future use. It is a traditional method of storing rain water in underground tanks, ponds, lakes, check dams and used in future.

The main purpose of rainwater harvesting is to make the rainwater percolate under the ground so as to recharge '**groundwater level**'.

Methods of rainwater harvesting

(i) Roof top rainwater harvesting: Roof-tops are excellent **rain catchers**. The rain water that falls on the roof of the houses, apartments, commercial buildings etc. is collected and stored in the surface tank and can be used for domestic purpose.

(ii) Recharge pit: In this method, the rainwater is first collected from the roof tops or open spaces and is directed into the **percolation pits** through pipes for filtration. After filtration the rainwater enters the **recharge pits** or **ground wells**.



Figure 22.3 Rain water Harvesting

People living in rural areas adopt a variety of water collecting methods to capture and store as rain water. Some of the methods used are

(i) Digging of tanks or lakes (Eris): It is one of the **traditional water harvesting system** in Tamil Nadu. Eris are constructed in such a way that if the water in one eri overflows, it automatically gets diverted to the eri of the next village, as these eris are interconnected.

(ii) Ooranis: These are **small ponds** to collect rainwater. The water is used for various domestic purposes (drinking, washing and bathing). These ponds cater the nearby villages.

More to Know

kallanai Dam, also known as Grand Anicut, is the fourth oldest dam in the world, constructed by King Karikala Chola of the Chola Dynasty in the 2nd century A.D.(CE). It still serves the people of Tamilnadu, The dam is located on the River Kaveri, approximately 20 km from the city of Tiruchirapalli.

10th_Science Unit-22.indd 322 11-01-2020 15:44:01